

# EGFR fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes EGFR gene fusions from the msk\_impact\_50k\_2026 study, covering 55 fusion events. 28/55 fusions (50.9%) were in-frame. 41/55 fusions (74.5%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ( $p=0.0114539$ , statistically significant at  $\alpha=0.05$ ). No literature benchmark is configured for EGFR.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 55 analyzed fusion events, identifying 31 distinct fusion partner genes. The most frequent partner was SEPTIN14, observed in 18 events.

### Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 25/28 (89.3%) of in-frame fusions retained the domain,  $p=0.0114539$  (statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.247525.

### Domain disruption

Domain-disruption analysis was skipped: EGFR has no disruption\_required\_domains configured; domain-disruption analysis was skipped.

### Cutpoint detection

Cutpoint detection scanned 55 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 982 aa (permutation-corrected  $p=0.128713$ , not statistically significant at  $\alpha=0.05$ ). This is 17 aa from the nearest configured domain boundary at 965 aa.

### Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: EGFR has no group\_field configured; confidence-stats analysis was skipped.

### Composite evidence score

Composite evidence score reported: domain\_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption\_required\_domains not configured for this gene).

### Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

## Joint partner

Joint partner reported: EGFR has no gene\_pair configured; joint-partner analysis was skipped.

# Tables

## composite\_score tables

### composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score  | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                    | Composite_score     | Rank |
|--------------|-------------|-------------------|------------------------|-------------------------|--------------------------|----------------------------|----------------------------------------------------------|---------------------|------|
| SEPTIN14     | 18          | 0.327272727272727 | 0.060638136511060495   | None                    | 0.9786998132586368       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.3716376061405412  | 1    |
| RAD51        | 2           | 0.036363636363636 | 0.060638136511060495   | None                    | 0.9989495798319328       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2513015171594942  | 2    |
| LANCL2       | 3           | 0.054545454545454 | 0.060638136511060495   | None                    | 0.9618347338935574       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.25114054367919303 | 3    |
| ELAPOR2      | 1           | 0.018181818181818 | 0.060638136511060495   | None                    | 0.9989495798319328       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2435093093672864  | 4    |
| PKD1L1       | 1           | 0.018181818181818 | 0.060638136511060495   | None                    | 0.9989495798319328       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2435093093672864  | 5    |
| VPS41        | 1           | 0.018181818181818 | 0.060638136511060495   | None                    | 0.9957983193277311       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2428340392592432  | 6    |
| ZNF713       | 1           | 0.018181818181818 | 0.060638136511060495   | None                    | 0.9401260504201681       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2309042673504797  | 7    |
| VWC2         | 2           | 0.036363636363636 | 0.060638136511060495   | None                    | 0.9028361344537815       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2307057788641761  | 8    |
| NIPSNAP2     | 3           | 0.054545454545454 | 0.060638136511060495   | None                    | 0.8658963585434174       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2305823203898773  | 9    |
| TNRC18       | 1           | 0.018181818181818 | 0.060638136511060495   | None                    | 0.9233193277310925       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2273028267742492  | 10   |
| BLTP3B       | 1           | 0.018181818181818 | 0.060638136511060495   | None                    | 0.9023109243697479       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.22280102605396107 | 11   |
| CDH7         | 1           | 0.018181818181818 | 0.060638136511060495   | None                    | 0.832983193277311        | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2079450836770103  | 12   |
| FADD         | 2           | 0.036363636363636 | 0.060638136511060495   | None                    | 0.6901260504201681       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.18512504657125892 | 13   |
| EEA1         | 1           | 0.018181818181818 | 0.060638136511060495   | None                    | 0.6911764705882353       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.17755792881506555 | 14   |

| Partner_gene | Event_count | Recurrence_score    | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                    | Composite_score      | Rank |
|--------------|-------------|---------------------|------------------------|-------------------------|--------------------------|----------------------------|----------------------------------------------------------|----------------------|------|
| KIF5B        | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.6911764705882353       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.17755792881506555  | 15   |
| VSTM2A       | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.6911764705882353       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.17755792881506555  | 16   |
| CADPS        | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.6901260504201681       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.17733283877905112  | 17   |
| NUMA1        | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.6901260504201681       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.17733283877905112  | 18   |
| PDE1C        | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.6523109243697479       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.16922959748253252  | 19   |
| YIF1B        | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.6407563025210083       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.16675360708637407  | 20   |
| TUT7         | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.6271008403361344       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.1638274366181868   | 21   |
| CSF2RA       | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.3214285714285714       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.09832623613799471  | 22   |
| DDC          | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.2804621848739496       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.08954772473343288  | 23   |
| GARS1        | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.2804621848739496       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.08954772473343288  | 24   |
| SEL1L        | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.2804621848739496       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.08954772473343288  | 25   |
| TNS3         | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.2804621848739496       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.08954772473343288  | 26   |
| PCDH15       | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.2710084033613446       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.08752191440930324  | 27   |
| VOPP1        | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.23004201680672265      | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.07874340300474139  | 28   |
| SCAF4        | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.11764705882352944      | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.05465876915119999  | 29   |
| ZBPB         | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.006302521008403339     | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.030799225333672974 | 30   |

| Partner_gene | Event_count | Recurrence_score    | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                    | Composite_score      | Rank |
|--------------|-------------|---------------------|------------------------|-------------------------|--------------------------|----------------------------|----------------------------------------------------------|----------------------|------|
| PL0D3        | 1           | 0.01818181818181818 | 0.060638136511060495   | None                    | 0.0                      | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.029448685117586543 | 31   |

## cutpoint\_detection tables

### cutpoint\_scan

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio          | p_value              | neg_log10_p_value   |
|----------|------------------------|------------------|------------------------|------------------|---------------------|----------------------|---------------------|
| 30       | 1                      | 40               | 0                      | 14               | None                | 1.0                  | -0.0                |
| 36       | 2                      | 39               | 0                      | 14               | None                | 1.0                  | -0.0                |
| 142      | 3                      | 38               | 0                      | 14               | None                | 0.5624166190203925   | 0.24994185469837954 |
| 249      | 3                      | 38               | 1                      | 13               | 1.0263157894736843  | 1.0                  | -0.0                |
| 288      | 3                      | 38               | 2                      | 12               | 0.47368421052631576 | 0.5924468510484049   | 0.22735060452753406 |
| 297      | 7                      | 34               | 2                      | 12               | 1.2352941176470589  | 1.0                  | -0.0                |
| 336      | 8                      | 33               | 2                      | 12               | 1.4545454545454546  | 1.0                  | -0.0                |
| 599      | 9                      | 32               | 2                      | 12               | 1.6875              | 0.7093311767774617   | 0.14915095146564436 |
| 627      | 9                      | 32               | 3                      | 11               | 1.03125             | 1.0                  | -0.0                |
| 640      | 10                     | 31               | 3                      | 11               | 1.1827956989247312  | 1.0                  | -0.0                |
| 651      | 11                     | 30               | 3                      | 11               | 1.3444444444444446  | 1.0                  | -0.0                |
| 687      | 13                     | 28               | 5                      | 9                | 0.8357142857142857  | 1.0                  | -0.0                |
| 688      | 15                     | 26               | 6                      | 8                | 0.7692307692307693  | 0.7548366596998944   | 0.12214701587495107 |
| 823      | 15                     | 26               | 7                      | 7                | 0.5769230769230769  | 0.528760430668828    | 0.27674105235972185 |
| 875      | 15                     | 26               | 9                      | 5                | 0.32051282051282054 | 0.1175979559898796   | 0.9296002268145651  |
| 981      | 15                     | 26               | 10                     | 4                | 0.23076923076923078 | 0.031971018585759146 | 1.4952435270336717  |
| 982      | 21                     | 20               | 12                     | 2                | 0.175               | 0.02895546314423952  | 1.5382694841107174  |
| 983      | 31                     | 10               | 13                     | 1                | 0.23846153846153847 | 0.25480492869908067  | 0.5937921756847577  |
| 986      | 32                     | 9                | 13                     | 1                | 0.27350427350427353 | 0.4229368950729017   | 0.37372442734387856 |
| 994      | 33                     | 8                | 13                     | 1                | 0.3173076923076923  | 0.4208384837822578   | 0.3758845527657329  |
| 1003     | 34                     | 7                | 13                     | 1                | 0.37362637362637363 | 0.663943317750199    | 0.17786899569082928 |
| 1039     | 35                     | 6                | 13                     | 1                | 0.44871794871794873 | 0.663943317750199    | 0.17786899569082928 |
| 1054     | 37                     | 4                | 13                     | 1                | 0.7115384615384616  | 1.0                  | -0.0                |
| 1055     | 38                     | 3                | 14                     | 0                | 0.0                 | 0.5624166190203926   | 0.24994185469837946 |
| 1060     | 39                     | 2                | 14                     | 0                | 0.0                 | 1.0                  | -0.0                |
| 1075     | 40                     | 1                | 14                     | 0                | 0.0                 | 1.0                  | -0.0                |

## domain\_retention tables

*frame\_domain\_contingency\_table*

|    |    |
|----|----|
| 25 | 16 |
| 3  | 11 |

*domain\_retention\_descriptives*

| Domain_key | Domain_name                                  | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|------------|----------------------------------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| kinase     | Protein tyrosine and serine/threonine kinase | 55                      | 41                   | 3               | 11               | 0.10304912478825523                             |

## frequency tables

*Partner\_gene\_counts*

| Partner_gene | Event_count |
|--------------|-------------|
| BLTP3B       | 1           |
| CADPS        | 1           |
| CDH7         | 1           |
| CSF2RA       | 1           |
| DDC          | 1           |
| EEA1         | 1           |
| ELAPOR2      | 1           |
| FADD         | 2           |
| GARS1        | 1           |
| KIF5B        | 1           |
| LANCL2       | 3           |
| NIPSNAP2     | 3           |
| NUMA1        | 1           |
| PCDH15       | 1           |
| PDE1C        | 1           |
| PKD1L1       | 1           |
| PLOD3        | 1           |
| RAD51        | 2           |
| SCAF4        | 1           |
| SEL1L        | 1           |
| SEPTIN14     | 18          |

| Partner_gene | Event_count |
|--------------|-------------|
| TNRC18       | 1           |
| TNS3         | 1           |
| TUT7         | 1           |
| VOPP1        | 1           |
| VPS41        | 1           |
| VSTM2A       | 1           |
| VWC2         | 2           |
| YIF1B        | 1           |
| ZNF713       | 1           |
| ZPBP         | 1           |

## Per-event results (results.tsv)

| event_id                        | sample_id                 | patient_id | fusion_name         | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                       | source_site2_effect_on_frame |
|---------------------------------|---------------------------|------------|---------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>1404-T01-I<br>M6-2 | P-0031<br>404-T0<br>1-IM6 |            | BLTP<br>3B-E<br>GFR | BLT<br>P3B   | unknown      | five_prime  | 552702<br>72       | 27              | 1075                        | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005228) - UHRF1BP1L (NM_015054) Rearrangement :t(7;12)(p11.2;q23.1)(chr7:g.55270272::chr12:g.100460508)   Protein Fusion: mid-exon {EGFR:UHRF1BP1L} | NA                           |
| EVT-P-005<br>0679-T02-I<br>M6-3 | P-0050<br>679-T0<br>2-IM6 |            | CADP<br>S-EG<br>FR  | CA<br>DPS    | out-of-frame | three_prime | 552410<br>99       | 17              | 687                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | CADPS (NM_003716) - EGFR (NM_005228) fusion: t(3;7)(p14.2;p11.2)(chr3:g.62537410::chr7:g.55241099)   Protein Fusion: out of frame {CADPS:EGFR}               | NA                           |
| EVT-P-003<br>8683-T01-I<br>M6-4 | P-0038<br>683-T0<br>1-IM6 |            | CSF2<br>RA-E<br>GFR | CSF<br>2RA   | unknown      | three_prime | 552242<br>20       | 9               | 336                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | CSF2RA (NM_001161530) - EGFR (NM_005228) rearrangement: t(7;X)(p11.2;p22.33)(chr7:g.55224220::chrX:g.1401662)   Protein Fusion: mid-exon {CSF2RA:EGFR}       | NA                           |
| EVT-P-001<br>6795-T01-I<br>M6-5 | P-0016<br>795-T0<br>1-IM6 |            | DDC-<br>EGFR        | DD<br>C      | out-of-frame | three_prime | 552222<br>04       | 7               | 297                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | DDC (NM_000790) - EGFR (NM_005228) Rearrangement : c.435+4059 :DDC_c.889+359:EGFRinv   Protein Fusion: out of frame {DDC:EGFR}                               | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name  | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                                 | disrupted_domains                            | source_annotation_text                                                                                                                                   | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|--------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|----------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>8389-T02-I<br>M6-6   | P-0018<br>389-T0<br>2-IM6 |            | EEA1-EGFR    | EEA1         | in-frame     | threonine_prime | 55241575           | 18              | 688                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                                              | EEA1 (NM_003566) - EGFR (NM_005228) rearrangement: t(7;12)(p11.2;q22)(chr7:g.55241575::chr12:g.932211089)   Protein Fusion: in frame {EEA1:EGFR}         | NA                            |
| EVT-P-006<br>4660-T01-I<br>M7-9   | P-0064<br>660-T0<br>1-IM7 |            | EGFR-CDH7    | CDH7         | out-of-frame | five_prime      | 55254206           | 20              | 823                         | True                   | disrupted     | 0.43873517786561267      | True                |                                              |                                              | Protein tyrosine and serine/threonine kinase | EGFR (NM_005228) - CDH7 (NM_033646) rearrangement: t(7;18)(p11.2;q22.1)(chr7:g.55254206::chr18:g.63486938)   Protein Fusion: out of frame {EGFR:CDH7}    | NA                            |
| EVT-P-003<br>4892-T01-I<br>M6-497 | P-0034<br>892-T0<br>1-IM6 |            | EGFR-ELAPOR2 | ELAPOR2      | unknown      | threonine_prime | 55268103           | 24              | 981                         | False                  | lost          | 0.0                      | False               |                                              | Protein tyrosine and serine/threonine kinase |                                              | EGFR (NM_005228) rearrangement: c.2943:EGFR_chr7:g.86587163del   Antisense Fusion                                                                        | NA                            |
| EVT-P-005<br>9222-T01-I<br>M7-500 | P-0059<br>222-T0<br>3-IM7 |            | EGFR-LANCL2  | LANCL2       | in-frame     | five_prime      | 55259704           | 21              | 875                         | True                   | disrupted     | 0.6442687747035574       | True                |                                              |                                              | Protein tyrosine and serine/threonine kinase | EGFR (NM_005228) - LANCL2 (NM_018697) rearrangement: c.2625+137:EGFR_c.1008+3812:LANCL2del   Protein Fusion: in frame {EGFR:LANCL2}                      | NA                            |
| EVT-P-005<br>7880-T01-I<br>M6-501 | P-0057<br>880-T0<br>1-IM6 |            | EGFR-LANCL2  | LANCL2       | in-frame     | five_prime      | 55268871           | 25              | 983                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                                              | EGFR (NM_005228) - LANCL2 (NM_018697) fusion: c.2947-10:EGFR_c.205-8690:LANCL2del   Protein Fusion: in frame {EGFR:LANCL2}                               | NA                            |
| EVT-P-006<br>3142-T01-I<br>M7-503 | P-0063<br>142-T0<br>1-IM7 |            | EGFR-PCDH15  | PCDH15       | unknown      | five_prime      | 55221820           | 7               | 288                         | False                  | lost          | 0.0                      | False               |                                              | Protein tyrosine and serine/threonine kinase |                                              | EGFR (NM_005228) - PCDH15 (NM_001142763) rearrangement: t(7;10)(p11.2;q21.1)(chr7:g.55221820::chr10:g.56187463)   Protein Fusion: mid-exon {EGFR:PCDH15} | NA                            |
| EVT-P-004<br>2236-T01-I<br>M6-504 | P-0042<br>236-T0<br>1-IM6 |            | EGFR-PKD1L1  | PKD1L1       | in-frame     | five_prime      | 55268824           | 25              | 983                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                                              | EGFR (NM_005228) - PKD1L1 (NM_138295) fusion: c.2947-57:EGFR_c.8527-7825:PKD1L1inv   Protein Fusion: in frame {EGFR:PKD1L1}                              | NA                            |



| event_id                          | sample_id                 | patient_id | fusion_name           | partner_gene     | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                 | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------------|------------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|--------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-004<br>1149-T04-I<br>M6-505 | P-0041<br>149-T0<br>4-IM6 |            | EGFR<br>-RAD<br>51    | RAD<br>D51       | in-frame     | five_prime  | 552687<br>99       | 25              | 983                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005228) - RAD51 (NM_002875) fusion: t(7;15)(p11.2;q15.1)(chr7:g.55268799::chr15:g.40996189)   Protein Fusion: in frame {EGFR:RAD51}                           | NA                            |
| EVT-P-004<br>1149-T01-I<br>M6-506 | P-0041<br>149-T0<br>1-IM6 |            | EGFR<br>-RAD<br>51    | RAD<br>D51       | in-frame     | five_prime  | 552687<br>99       | 25              | 983                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005228) - RAD51 (NM_002875) fusion: t(7;15)(p11.2;q15.1)(chr7:g.55268799::chr15:g.40996189)   Protein Fusion: in frame {EGFR:RAD51}                           | NA                            |
| EVT-P-003<br>4594-T01-I<br>M6-507 | P-0034<br>594-T0<br>1-IM6 |            | EGFR<br>-SEP<br>TIN14 | SEP<br>TIN<br>14 | in-frame     | five_prime  | 552685<br>44       | 25              | 983                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-110:EGFR_c.1120-1204:SEPT14   Protein Fusion: in frame {EGFR:SEPT14}                                              | NA                            |
| EVT-P-003<br>8138-T01-I<br>M6-508 | P-0038<br>138-T0<br>1-IM6 |            | EGFR<br>-SEP<br>TIN14 | SEP<br>TIN<br>14 | in-frame     | five_prime  | 552683<br>81       | 24              | 982                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2946+275:EGFR_c.1119+2043:SEPT14inv   Protein Fusion: in frame {EGFR:SEPT14} | NA                            |
| EVT-P-003<br>7526-T01-I<br>M6-509 | P-0037<br>526-T0<br>1-IM6 |            | EGFR<br>-SEP<br>TIN14 | SEP<br>TIN<br>14 | unknown      | five_prime  | 552689<br>16       | 25              | 994                         | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2982:EGFR_c.1119+1151:SEPT14inv   Protein Fusion: mid-exon {EGFR:SEPT14}                                               | NA                            |
| EVT-P-003<br>9255-T02-I<br>M6-510 | P-0039<br>255-T0<br>2-IM6 |            | EGFR<br>-SEP<br>TIN14 | SEP<br>TIN<br>14 | in-frame     | five_prime  | 552708<br>07       | 27              | 1091                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.3271+489:EGFR_c.818-3507:SEPT14inv   Protein Fusion: in frame {EGFR:SEPT14}                                            | NA                            |
| EVT-P-003<br>2604-T01-I<br>M6-512 | P-0032<br>604-T0<br>1-IM6 |            | EGFR<br>-SEP<br>TIN14 | SEP<br>TIN<br>14 | in-frame     | five_prime  | 552687<br>71       | 25              | 983                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-110:EGFR_c.1120-1204:SEPT14   Protein Fusion: in frame {EGFR:SEPT14}                                              | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name       | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                                 | disrupted_domains | source_annotation_text                                                                                                                                                 | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|----------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-003<br>1753-T01-I<br>M6-513 | P-0031<br>753-T0<br>1-IM6 |            | EGFR<br>-SEPTIN14 | SEPTIN14     | in-frame     | five_prime  | 55268412           | 24              | 982                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2946+306:EGFR_c.1119+2531:SEPT14inv   Protein Fusion: in frame {EGFR:SEPT14} | NA                           |
| EVT-P-000<br>5701-T01-I<br>M5-514 | P-0005<br>701-T0<br>1-IM5 |            | EGFR<br>-SEPTIN14 | SEPTIN14     | in-frame     | five_prime  | 55269490           | 26              | 1054                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) rearrangement : c.3162+15:EGFR_c.1119+2833:SEPT14inv   Protein fusion: in frame {EGFR:SEPT14}                                    | NA                           |
| EVT-P-004<br>3525-T01-I<br>M6-515 | P-0043<br>525-T0<br>1-IM6 |            | EGFR<br>-SEPTIN14 | SEPTIN14     | in-frame     | five_prime  | 55268859           | 25              | 983                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-22:EGFR_c.1120-807:SEPT14inv   Protein Fusion: in frame {EGFR:SEPT14}                                             | NA                           |
| EVT-P-002<br>9159-T01-I<br>M6-516 | P-0029<br>159-T0<br>1-IM6 |            | EGFR<br>-SEPTIN14 | SEPTIN14     | in-frame     | three_prime | 55269850           | 27              | 1055                        | True                   | lost          | 0.0                      | False               |                                              | Protein tyrosine and serine/threonine kinase |                   | SEPT14 (NM_207366) - EGFR (NM_005228) rearrangement: c.1119+1369:SEPT14_c.3163-360:EGFRinv   Protein Fusion: in frame {SEPT14:EGFR}                                    | NA                           |
| EVT-P-002<br>8241-T01-I<br>M6-517 | P-0028<br>241-T0<br>1-IM6 |            | EGFR<br>-SEPTIN14 | SEPTIN14     | in-frame     | five_prime  | 55268525           | 25              | 983                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2947-356:EGFR_c.1119+4312:SEPT14inv   Protein Fusion: in frame {EGFR:SEPT14} | NA                           |
| EVT-P-002<br>5191-T02-I<br>M6-518 | P-0025<br>191-T0<br>2-IM6 |            | EGFR<br>-SEPTIN14 | SEPTIN14     | in-frame     | five_prime  | 55269542           | 26              | 1054                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.3162+67:EGFR_c.1120-3396:SEPT14inv   Protein Fusion: in frame {EGFR:SEPT14}                                            | NA                           |
| EVT-P-006<br>1245-T01-I<br>M7-519 | P-0061<br>245-T0<br>1-IM7 |            | EGFR<br>-SEPTIN14 | SEPTIN14     | in-frame     | five_prime  | 55268815           | 25              | 983                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) rearrangement: c.2947-66:EGFR_c.1119+2973:SEPT14inv   Protein Fusion: in frame {EGFR:SEPT14}                                     | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name       | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                    | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-001<br>4220-T02-I<br>M6-520 | P-0014<br>220-T0<br>2-IM6 |            | EGFR-<br>SEPTIN14 | SEPTIN14     | unknown      | five_prime  | 55268942           | 25              | 1003                        | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-25 fused with SEPT14 exon 10): c.3008:EGFR_c.1120-4102:SEPT14inv   Protein Fusion: mid-exon {EGFR:SEPT14}      | NA                            |
| EVT-P-002<br>3043-T01-I<br>M6-521 | P-0023<br>043-T0<br>1-IM6 |            | EGFR-<br>SEPTIN14 | SEPTIN14     | in-frame     | five_prime  | 55268179           | 24              | 982                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) Fusion (EGFR exons 1-24 fused with SEPT14 exon 10) : c.2946+73:EGFR_c.1120-180:SEPT14   Protein Fusion: in frame {EGFR:SEPT14}      | NA                            |
| EVT-P-001<br>9603-T01-I<br>M6-522 | P-0019<br>603-T0<br>1-IM6 |            | EGFR-<br>SEPTIN14 | SEPTIN14     | in-frame     | five_prime  | 55268171           | 24              | 982                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) Fusion ((EGFR exons 1-24 fused with SEPT14 exon 10) : c.2946+65:EGFR_c.1119+3306:SEPT14   Protein Fusion: in frame {EGFR:SEPT14}    | NA                            |
| EVT-P-004<br>7356-T01-I<br>M6-523 | P-0047<br>356-T0<br>1-IM6 |            | EGFR-<br>SEPTIN14 | SEPTIN14     | in-frame     | five_prime  | 55268316           | 24              | 982                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2946+210:EGFR_c.1120-2817:SEPT14inv   Protein Fusion: in frame {EGFR:SEPT14}                                              | NA                            |
| EVT-P-001<br>6256-T01-I<br>M6-524 | P-0016<br>256-T0<br>1-IM6 |            | EGFR-<br>SEPTIN14 | SEPTIN14     | in-frame     | five_prime  | 55268390           | 24              | 982                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005228) - SEPT14 (NM_207366) fusion( EGFR exons 1-24 fused to SEPT14 exons 9-10): c.2946+284:EGFR_c.1120-3940:SEPT14inv   Protein Fusion: in frame {EGFR:SEPT14} | NA                            |
| EVT-P-001<br>5179-T01-I<br>M6-525 | P-0015<br>179-T0<br>1-IM6 |            | EGFR-<br>SEPTIN14 | SEPTIN14     | in-frame     | five_prime  | 55268650           | 25              | 983                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005229) - SEPT14 (NM_014857) fusion (EGFR exons 1-24 fused with SEPT14 exon 10): c.2947-231:EGFR_c.1119+3910:SEPT14inv   Protein fusion: in frame (EGFR-SEPT14)  | NA                            |
| EVT-P-005<br>8375-T01-I<br>M6-526 | P-0058<br>375-T0<br>1-IM6 |            | EGFR-<br>TNRC18   | TNRC18       | out-of-frame | five_prime  | 55269897           | 27              | 1055                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                   | EGFR (NM_005228) - TNRC18 (NM_001080495) fusion: c.3163-313:EGFR_c.5720-6144:TNRC18inv   Protein Fusion: out of frame {EGFR:TNRC18}                                       | NA                            |

| event_id                          | sample_id                 | patient_id | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                                 | disrupted_domains | source_annotation_text                                                                                                                          | source_site2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|----------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-001<br>6009-T01-I<br>M6-527 | P-0016<br>009-T0<br>1-IM6 |            | EGFR<br>-VOPP1  | VOPP1        | in-frame     | five_prime  | 55220393           | 6               | 249                         | True                   | lost          | 0.0                      | False               |                                              | Protein tyrosine and serine/threonine kinase |                   | EGFR (NM_005228) - VOPP1 (NM_030796): c.747+36:EGFR_c.55-23161VOPP1inv   Protein Fusion: in frame {EGFR:VOPP1}                                  | NA                           |
| EVT-P-003<br>9262-T01-I<br>M6-529 | P-0039<br>262-T0<br>1-IM6 |            | EGFR<br>-VSTM2A | VSTM2A       | out-of-frame | five_prime  | 55241509           | 18              | 688                         | True                   | lost          | 0.0                      | False               |                                              | Protein tyrosine and serine/threonine kinase |                   | EGFR (NM_005228) - VSTM2A (NM_182546) rearrangement: c.2062-105:EGFR_c.634+2190:VSTM2A dup   Protein Fusion: out of frame {EGFR:VSTM2A}         | NA                           |
| EVT-P-004<br>2236-T01-I<br>M6-530 | P-0042<br>236-T0<br>1-IM6 |            | EGFR<br>-VWC2   | VWC2         | unknown      | five_prime  | 55270225           | 27              | 1060                        | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                   | EGFR (NM_005228) - VWC2 (NM_198570) fusion: c.3178:EGFR_c.827-13914:VWC2dup   Protein Fusion: mid-exon {EGFR:VWC2}                              | NA                           |
| EVT-P-004<br>3097-T01-I<br>M6-531 | P-0043<br>097-T0<br>1-IM6 |            | EGFR<br>-ZNF713 | ZNF713       | in-frame     | five_prime  | 55269400           | 26              | 1039                        | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                   | EGFR (NM_005228) - ZNF713 (NM_182633) fusion: c.3115-28:EGFR_c.49-962:ZNF713del   Protein Fusion: in frame {EGFR:ZNF713}                        | NA                           |
| EVT-P-002<br>7944-T01-I<br>M6-534 | P-0027<br>944-T0<br>1-IM6 |            | FADD<br>-EGFR   | FADD         | unknown      | five_prime  | 55241077           | 17              | 687                         | True                   | lost          | 0.0                      | False               |                                              | Protein tyrosine and serine/threonine kinase |                   | EGFR (NM_005228) - FADD (NM_003824) rearrangement: t(7;11)(p11.2;q13.3)(chr7:g55241077::chr11:g70052330)   Protein Fusion: mid-exon {EGFR:FADD} | NA                           |
| EVT-P-002<br>7944-T03-I<br>M6-535 | P-0027<br>944-T0<br>3-IM6 |            | FADD<br>-EGFR   | FADD         | unknown      | five_prime  | 55241077           | 17              | 687                         | True                   | lost          | 0.0                      | False               |                                              | Protein tyrosine and serine/threonine kinase |                   | EGFR (NM_005228) - FADD (NM_003824) rearrangement: t(7;11)(p11.2;q13.3)(chr7:g55241077::chr11:g70052330)   Protein Fusion: mid-exon {EGFR:FADD} | NA                           |

| event_id                          | sample_id                 | patient_id | fusion_name    | partner_gene | frame_status | target_role      | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                                         | lost_domains                                             | disrupted_domains | source_annotation_text                                                                                                                               | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|----------------|--------------|--------------|------------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------------------|----------------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-006<br>3743-T01-I<br>M7-536 | P-0063<br>743-T0<br>1-IM7 |            | GARS<br>1-EGFR | GARS1        | out-of-frame | threep<br>_prime | 552223<br>68       | 7               | 297                         | True                   | retained      | 1.0                      | False               | Protein<br>tyrosine<br>and<br>serine/threonine<br>kinase |                                                          |                   | GARS (NM_002047) - EGFR (NM_005228) fusion: c.1032-338:GARS_c.889+523:EGFRdel   Protein Fusion: out of frame {GARS:EGFR}                             | NA                            |
| EVT-P-002<br>1038-T01-I<br>M6-538 | P-0021<br>038-T0<br>1-IM6 |            | KIF5B<br>-EGFR | KIF5B        | in-frame     | threep<br>_prime | 552414<br>86       | 18              | 688                         | True                   | retained      | 1.0                      | False               | Protein<br>tyrosine<br>and<br>serine/threonine<br>kinase |                                                          |                   | KIF5B (NM_004521) - EGFR (NM_005228) rearrangement: t(7;10)(p11.2;q11.22)(chr7:g.55241486::chr10:g.32312089)   Protein Fusion: in frame {KIF5B:EGFR} | NA                            |
| EVT-P-002<br>5018-T01-I<br>M6-539 | P-0025<br>018-T0<br>1-IM6 |            | LANCL2-EGFR    | LANCL2       | out-of-frame | threep<br>_prime | 552685<br>37       | 25              | 983                         | True                   | lost          | 0.0                      | False               |                                                          | Protein<br>tyrosine<br>and<br>serine/threonine<br>kinase |                   | LANCL2 (NM_018697) - EGFR (NM_005228) rearrangement: c.1259-255:LANCL2_c.2947-344:EGFRdup   Protein Fusion: out of frame {LANCL2:EGFR}               | NA                            |
| EVT-P-005<br>9222-T01-I<br>M7-540 | P-0059<br>222-T0<br>1-IM7 |            | NIPSNAP2-EGFR  | NIPSNAP2     | unknown      | threep<br>_prime | 552330<br>46       | 15              | 599                         | False                  | retained      | 1.0                      | False               | Protein<br>tyrosine<br>and<br>serine/threonine<br>kinase |                                                          |                   | GBAS (NM_001483) - EGFR (NM_005228) fusion: c.92+717:GBAS_c.1796:EGFRdup   Protein Fusion: mid-exon {GBAS:EGFR}                                      | NA                            |
| EVT-P-003<br>7759-T01-I<br>M6-541 | P-0037<br>759-T0<br>1-IM6 |            | NIPSNAP2-EGFR  | NIPSNAP2     | out-of-frame | threep<br>_prime | 552684<br>44       | 24              | 982                         | True                   | lost          | 0.0                      | False               |                                                          | Protein<br>tyrosine<br>and<br>serine/threonine<br>kinase |                   | GBAS (NM_001483) - EGFR (NM_005228) Rearrangement: c.2946+338:EGFR_c.279-1159:GBASdup   Protein Fusion: out of frame {GBAS:EGFR}                     | NA                            |
| EVT-P-003<br>7759-T03-I<br>M6-542 | P-0037<br>759-T0<br>3-IM6 |            | NIPSNAP2-EGFR  | NIPSNAP2     | out-of-frame | threep<br>_prime | 552684<br>44       | 24              | 982                         | True                   | lost          | 0.0                      | False               |                                                          | Protein<br>tyrosine<br>and<br>serine/threonine<br>kinase |                   | GBAS (NM_001483) - EGFR (NM_005228) rearrangement: c.279-1159:GBAS_c.2946+338:EGFRdup   Protein Fusion: out of frame {GBAS:EGFR}                     | NA                            |

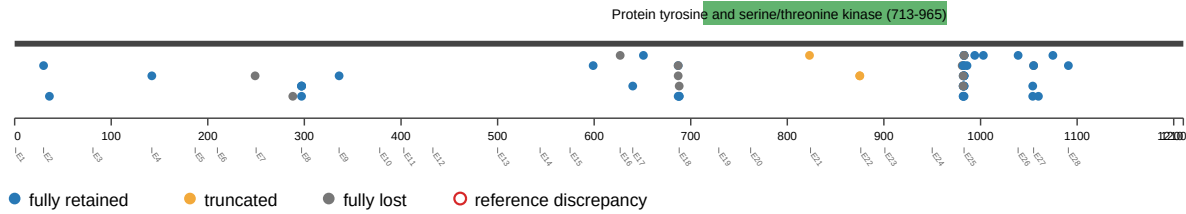
| event_id                          | sample_id                 | patient_id | fusion_name | partner_gene | frame_status | target_role     | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains                                 | disrupted_domains | source_annotation_text                                                                                                                                        | source_site_2_effect_on_frame |
|-----------------------------------|---------------------------|------------|-------------|--------------|--------------|-----------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|----------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-003<br>6028-T01-I<br>M6-543 | P-0036<br>028-T0<br>1-IM6 |            | NUMA1-EGFR  | NUMA1        | in-frame     | threonine_prime | 55241128           | 17              | 687                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                   | NUMA1 (NM_006185) - EGFR (NM_005228) rearrangement: t(7;11)(p11.2;q13.4)(chr7:g.55241128::chr11:g.71728415)   Protein Fusion: in frame {NUMA1:EGFR}           | NA                            |
| EVT-P-006<br>5470-T01-I<br>M7-545 | P-0065<br>470-T0<br>1-IM7 |            | PDE1C-EGFR  | PDE1C        | unknown      | threonine_prime | 55240707           | 17              | 651                         | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                   | PDE1C (NM_001191058) - EGFR (NM_005228) fusion: c.86-15224:PDE1C_c.1951:EGFRinv   Protein Fusion: mid-exon {PDE1C:EGFR}                                       | NA                            |
| EVT-P-005<br>9032-T01-I<br>M7-546 | P-0059<br>032-T0<br>1-IM7 |            | PLOD3-EGFR  | PLOD3        | out-of-frame | threonine_prime | 55179084           | 2               | 30                          | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                   | PLOD3 (NM_001084) - EGFR (NM_005228) fusion: c.2062-9:PLOD3_c.89-30895:EGFRinv   Protein Fusion: out of frame {PLOD3:EGFR}                                    | NA                            |
| EVT-P-002<br>8484-T02-I<br>M6-548 | P-0028<br>484-T0<br>2-IM6 |            | SCAF4-EGFR  | SCAF4        | out-of-frame | threonine_prime | 55214145           | 4               | 142                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                   | SCAF4 (NM_020706) - EGFR (NM_005228) fusion: t(7;21)(p11.2;q22.11)(chr7:g.55214145::chr21:g.33094864)   Protein Fusion: out of frame {SCAF4:EGFR}             | NA                            |
| EVT-P-003<br>0910-T01-I<br>M6-551 | P-0030<br>910-T0<br>1-IM6 |            | SEL1L-EGFR  | SEL1L        | in-frame     | threonine_prime | 55222485           | 7               | 297                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                   | SEL1L (NM_005065) - EGFR (NM_005228) rearrangement: t(7;14)(p11.2;q31.1)(chr7:g.55222485::chr14:g.81986576)   Protein Fusion: in frame {SEL1L:EGFR}           | NA                            |
| EVT-P-005<br>2976-T02-I<br>M7-552 | P-0052<br>976-T0<br>2-IM7 |            | TNS3-EGFR   | TNS3         | in-frame     | threonine_prime | 55222307           | 7               | 297                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |                                              |                   | TNS3 (NM_022748) - EGFR (NM_005228) fusion: c.1025-2026:TNS3_c.889+462:EGFRinv   Protein Fusion: in frame {TNS3:EGFR}                                         | NA                            |
| EVT-P-000<br>6873-T01-I<br>M5-554 | P-0006<br>873-T0<br>1-IM5 |            | TUT7-EGFR   | TUT7         | unknown      | five_prime      | 55238031           | 16              | 627                         | True                   | lost          | 0.0                      | False               |                                              | Protein tyrosine and serine/threonine kinase |                   | EGFR (NM_005228) - ZCCHC6 (NM_024617) rearrangement : t(7;9)(chr7p11.2;chr9q21.33)(chr7:g.55238031::chr9:g.88930920)   Protein fusion: mid-exon (EGFR-ZCCHC6) | NA                            |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains                             | lost_domains | disrupted_domains                            | source_annotation_text                                                                                                              | source_site_2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|--------------------------|---------------------|----------------------------------------------|--------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EVT-P-0015419-T01-IM6-555 | P-0015419-T01-IM6 |            | VPS41-EGFR  | VPS41        | unknown      | five_prime  | 55268890           | 25              | 986                         | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                                              | EGFR (NM_005228)- VPS41 (NM_014396)<br>Rearrangement : c.2956:EGFR_c.247-4922:VPS41inv   Protein fusion: mid-exon (EGFR-VPS41)      | NA                            |
| EVT-P-0030222-T01-IM6-557 | P-0030222-T01-IM6 |            | VWC2-EGFR   | VWC2         | out-of-frame | three_prime | 55259621           | 21              | 875                         | True                   | disrupted     | 0.35968379446640314      | True                |                                              |              | Protein tyrosine and serine/threonine kinase | VWC2 (NM_198570) - EGFR (NM_005228)<br>rearrangement: c.827-37789:VWC2_c.2625+54:EGFRdel   Protein Fusion: out of frame {VWC2:EGFR} | NA                            |
| EVT-P-0019922-T01-IM6-558 | P-0019922-T01-IM6 |            | YIF1B-EGFR  | YIF1B        | unknown      | three_prime | 55238939           | 16              | 640                         | True                   | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                                              | EGFR (NM_005228)<br>rearrangement: t(7;19)(p22.1;q13.2)(chr7:g.5523893::chr19:g.38806683)   Antisense Fusion                        | NA                            |
| EVT-P-0030222-T01-IM6-559 | P-0030222-T01-IM6 |            | ZBPB-EGFR   | ZBPB         | unknown      | three_prime | 55209996           | 2               | 36                          | False                  | retained      | 1.0                      | False               | Protein tyrosine and serine/threonine kinase |              |                                              | ZBPB (NM_007009) - EGFR (NM_005228)<br>rearrangement: c.487+5719:ZBPB_c.106:EGFRinv   Protein Fusion: mid-exon {ZBPB:EGFR}          | NA                            |

# Figures

## domain retention outliers

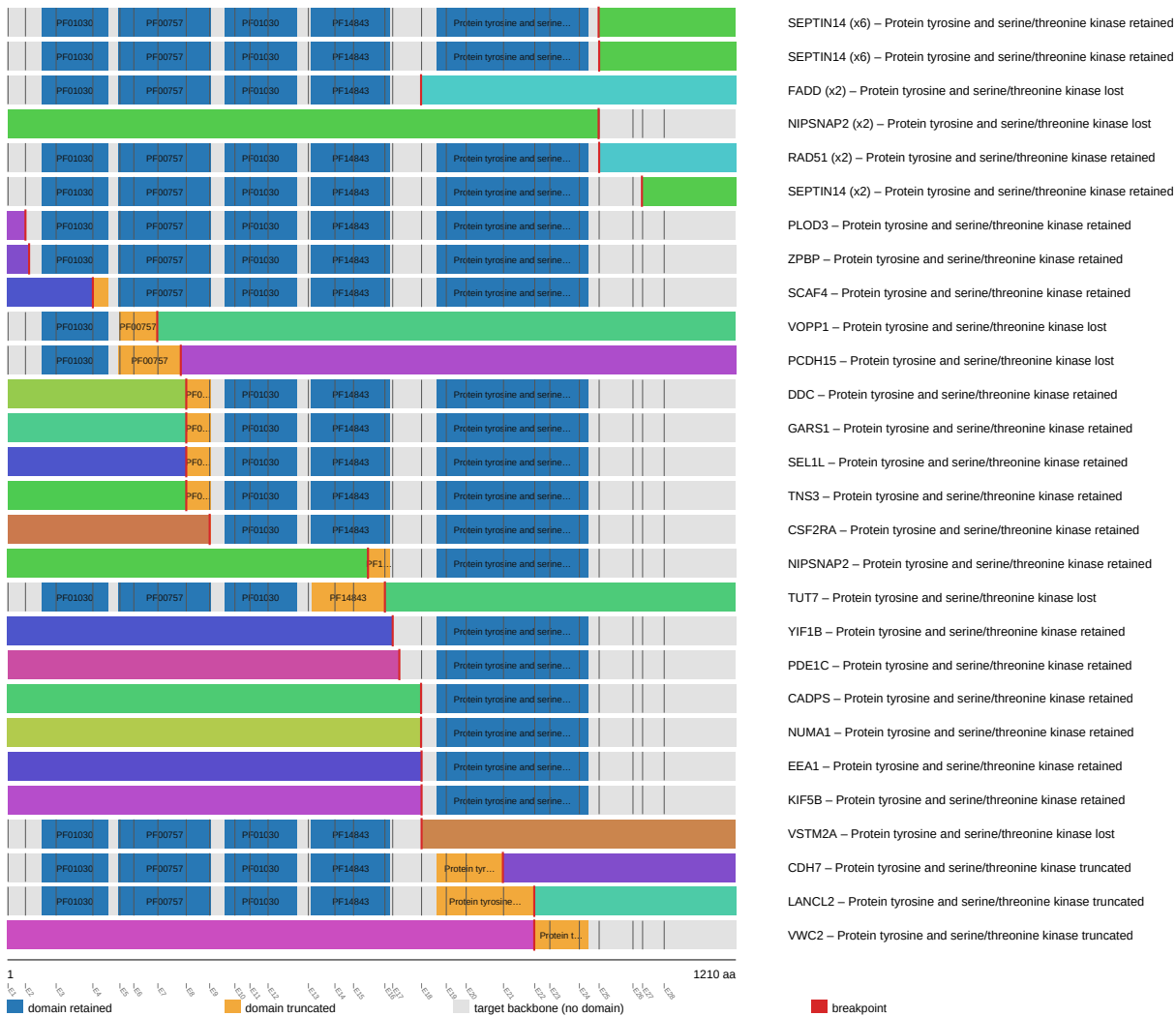
EGFR domain-retention track



## fusion schematic

EGFR fusion-transcript schematic

partner block → breakpoint → retained EGFR portion (domain-colored, exon ticks)





# intragenic deletion schematic

## EGFR intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



- 6-exon deletion (in-frame) (x279)
- 3-exon deletion (out-of-frame) (x24)
- 3-exon deletion (out-of-frame) (x19)
- 2-exon deletion (in-frame) (x5)
- 2-exon deletion (in-frame) (x5)
- 1-exon deletion (out-of-frame) (x5)
- 2-exon deletion (in-frame) (x3)
- 1-exon deletion (out-of-frame) (x3)
- 3-exon deletion (in-frame) (x2)
- 6-exon deletion (out-of-frame) (x2)
- 4-exon deletion (in-frame)
- 2-exon deletion (in-frame)
- 5-exon deletion (in-frame)
- 3-exon deletion (in-frame)
- 3-exon deletion (out-of-frame)
- 2-exon deletion (in-frame)
- 2-exon deletion (in-frame)

# reference comparison

## Reference vs msk\_impact\_50k\_2026

|                 |             |                |
|-----------------|-------------|----------------|
| In-frame        | <div></div> | reference 0.0% |
|                 | <div></div> | run 50.9%      |
| Domain retained | <div></div> | reference 0.0% |
|                 | <div></div> | run 74.5%      |